

DANIELS HARBOUR, NL, Canada

WMO: 711850

Lat: 50.2364N

Lon: 57.5811W

Elev: 19

StdP: 101.10

Time Zone: -3.50 (NAF)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.2	-16.1	-23.6	0.4	-17.6	-21.1	0.6	-15.5	18.6	-5.7	16.2	-4.2	5.2	290	0.808

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.6	21.5	17.7	20.3	17.1	19.3	16.5	18.6	20.6	17.7	19.6	17.0	18.8	6.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	12.8	19.7	16.9	12.1	18.9	16.0	11.4	18.3	53.1	20.8	50.3	19.7	47.9	18.9	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.8	13.1	11.7	DB	-22.7	25.6	3.5	1.7	-25.2	26.8	-27.3	27.8	-29.2	28.7	-31.8	30.0	(4)
(5)			WB	-23.5	20.7	3.6	1.0	-26.1	21.4	-28.2	22.0	-30.3	22.5	-32.9	23.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-6.3	-7.1	-4.1	1.0	6.1	10.7	15.2	15.9	12.1	6.8	2.0	-2.9	(6)	
	DBStd	8.77	4.90	5.15	5.01	3.63	3.12	3.22	2.60	2.46	3.38	3.37	4.09	4.38	(7)	
	HDD10.0	2620	506	478	438	271	128	29	1	1	16	110	243	399	(8)	
	HDD18.3	5176	764	712	696	520	380	230	101	81	187	357	490	658	(9)	
	CDD10.0	496	0	0	0	1	6	50	162	183	81	12	2	0	(10)	
	CDD18.3	9	0	0	0	0	0	0	3	4	2	0	0	0	(11)	
	CDH23.3	10	0	0	0	0	0	1	4	3	2	0	0	0	(12)	
	CDH26.7	1	0	0	0	0	0	0	0	1	0	0	0	0	(13)	
	Wind	WSAvg	5.7	6.5	6.0	5.8	5.8	5.0	5.0	5.1	5.0	5.7	5.7	6.1	6.3	(14)
(15) Precipitation	PrecAvg	1164	108	88	86	60	76	101	99	124	96	110	110	108	(15)	
	PrecMax	1480	174	198	169	140	155	302	211	230	172	202	245	210	(16)	
	PrecMin	728	37	12	20	6	18	36	29	34	35	50	32	22	(17)	
	PrecStd	201	37	42	42	34	35	48	44	48	35	39	45	43	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	8.8	9.7	14.3	17.2	20.7	23.5	23.0	23.0	18.2	16.8	10.4	(19)	
		MCWB	8.9	6.9	6.4	10.2	12.2	16.0	18.7	18.3	18.9	13.5	13.3	9.0	(20)	
	2%	DB	4.7	4.1	6.0	10.9	14.5	18.6	21.3	21.4	20.0	15.1	13.1	6.6	(21)	
		MCWB	3.3	2.4	3.7	7.5	10.7	14.9	17.6	17.8	17.0	12.6	10.8	5.4	(22)	
	5%	DB	2.2	1.5	3.5	8.3	12.8	17.0	20.0	20.4	18.2	13.4	9.8	4.4	(23)	
		MCWB	1.0	0.0	1.6	5.5	9.5	13.9	16.8	17.4	15.9	11.3	8.2	3.2	(24)	
	10%	DB	0.4	0.0	1.8	5.9	11.0	15.5	18.8	19.6	17.0	11.8	7.4	2.7	(25)	
		MCWB	-0.6	-1.2	-0.1	3.3	8.2	12.8	16.2	16.9	14.8	9.8	6.0	1.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	6.9	7.5	11.0	13.2	17.5	20.1	19.7	19.1	15.0	13.9	8.9	(27)	
		MCDB	10.4	8.5	9.5	13.4	15.8	19.4	21.7	21.7	22.4	17.3	16.3	10.3	(28)	
	2%	WB	3.5	2.5	4.1	7.7	11.5	15.8	18.5	18.7	17.5	13.0	11.0	5.6	(29)	
		MCDB	4.5	3.8	6.2	10.3	13.9	17.7	20.5	20.4	19.7	14.7	13.0	6.5	(30)	
	5%	WB	1.0	0.3	1.9	5.4	9.9	14.4	17.4	18.0	16.2	11.6	8.4	3.3	(31)	
		MCDB	1.9	1.2	3.5	7.9	12.1	16.5	19.2	19.8	17.9	13.2	9.7	4.1	(32)	
	10%	WB	-0.9	-1.2	0.2	3.6	8.3	13.2	16.6	17.2	15.1	10.0	6.2	1.5	(33)	
		MCDB	0.2	0.0	1.6	5.7	10.6	15.2	18.4	19.1	16.8	11.7	7.5	2.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	7.0	6.6	5.8	6.6	6.3	5.5	5.6	5.8	5.4	5.4	5.5	(35)	
		MCDBR	9.3	8.8	8.9	9.6	9.6	8.4	7.0	6.3	7.1	8.1	9.8	8.7	(36)	
		MCWBR	9.1	8.2	7.6	7.6	6.8	5.7	4.8	4.5	5.5	6.7	8.9	8.6	(37)	
	5% WB	MCDBR	9.2	9.4	8.8	9.4	9.2	7.5	6.4	5.6	6.7	7.7	9.6	8.6	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.295	0.310	0.318	0.350	0.372	0.370	0.410	0.379	0.360	0.336	0.307	0.283	(40)	
		taud	2.293	2.240	2.281	2.329	2.374	2.399	2.345	2.430	2.462	2.485	2.467	2.393	(41)	
	Ebn at Noon	Ebn at Noon	733	818	885	893	885	887	843	855	834	792	731	705	(42)	
		Edh at Noon	69	93	107	115	115	113	118	103	90	74	59	54	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.77	1.43	2.60	3.84	4.89	5.13	5.08	4.29	3.12	1.89	0.94	0.60	(44)	
	RadStd	RadStd	0.09	0.19	0.32	0.46	0.40	0.53	0.43	0.28	0.28	0.15	0.10	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (13 neighbors)	N/A	N/A	+1.28	N/A	+0.38	+0.50	N/A	N/A	N/A	N/A

Nomenclature: See separate page